Characterisation of Compton Scattering Geometry and Its Contribution to Timing Resolution in Time-of-Flight Detection Systems

RadiaTED

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ABSTRACT

Compton scattering is the dominant photon interaction in the energy range relevant to radiotherapy and PET imaging. When a high-energy photon interacts with a loosely bound electron, the electron is ejected and a lower-energy photon is scattered at an angle governed by relativistic kinematics. In time-of-flight (TOF) detection systems, the geometric variability of scattered-particle trajectories introduces an irreducible broadening of the timing distribution, which limits the precision with which annihilation vertices can be localised. This experiment aims to quantify, for the first time in a student-led context, the relative contributions of geometric path-length variation and intrinsic detector jitter to the integral timing resolution of a Compton-based TOF setup. Using an electron beam at CERN, a Bremsstrahlung photon source, a phantom Compton target, two plastic scintillators for timestamping, and delay wire chambers (DWC) for track reconstruction, we will measure the time difference Δt between two scintillator signals and correlate it with the reconstructed scattering angle and path length. By disentangling the geometric and detector-induced components of the timing distribution, this work is expected to provide experimentally grounded data relevant to the optimisation of next-generation TOF-PET systems.

Keywords: *Compton scattering, time-of-flight, timing resolution, PET imaging, delay wire chamber, Bremsstrahlung, plastic scintillator*

1. INTRODUCTION

The Compton effect occurs when a high-frequency photon interacts with a loosely bound charged particle, such as an electron, causing it to scatter. When a photon interacts with a valence electron of an atom or molecule, the electron is freed and a lower-energy scattered photon is emitted in a different direction.

Compton scattering is integral to radiobiology, as it is the most probable interaction between a gamma ray or X-ray used in radiotherapy and the tissue of an organism. Between 70 and 80% of the energy of X-rays used in radiotherapy undergoes Compton scattering. The electrons released by the Compton effect ionise the molecules along their path, damaging the DNA of cancerous cells. Damaged DNA disrupts metabolic activity and cell reproduction, thereby halting the spread of cancer.

In this experimental setup, we investigate the behaviour of the electrons produced in Compton scattering with the goal of better understanding radiation interactions in cancer treatments. Specifically, we aim to achieve a higher success rate in damaging cancer cells while minimising collateral damage to healthy tissue.

Building on this fundamental interaction, our experiment investigates how Compton-induced scattering influences timing measurements in time-of-flight-based detection systems. Specifically, we aim to calculate the integral timing resolution of a Compton-based setup by separating geometric path-length contributions from detector-induced time spread through track-assisted event reconstruction. By correlating scattering angle, energy deposition, and measured time differences, we analyse how Compton kinematics affect the effective spatial uncertainty in TOF systems.

While the kinematics of Compton scattering are well-established, no prior experiment has directly separated the geometric path-length contribution to timing resolution from intrinsic detector jitter using simultaneous track reconstruction and timestamping within the same event; this measurement addresses that gap.

1.1 Physics Background

In time-of-flight systems, knowledge of the event location is directly connected to time resolution. Time resolution measures how accurately a detector can determine the timing of events and how well it can distinguish between two closely spaced events. This connection is expressed by the equation $\Delta x = \frac{c\Delta t}{2}$, where Δt is the system's time resolution and Δx is the corresponding spatial uncertainty.

In PET, and especially in TOF-PET systems, improved timing resolution enables more precise localisation of annihilation events, as demonstrated by the directly proportional improvement in the signal-to-noise ratio. However, electronic noise is not the only factor limiting time resolution. Variation in particle trajectory lengths, interaction-vertex uncertainty, and multiple scatterings can all cause the timing distribution to broaden.

1.2 Research Question

The central question of this experiment is: to what extent do the scattering geometry and the variation in actual geometric path length contribute to the integral timing resolution of the system, independently of intrinsic detector jitter?

To answer this, both the timing and the spatial location of the event products must be measured simultaneously, and the two contributions must be disentangled through track-assisted reconstruction.

1.3 Note on Energy Regime

In clinical TOF-PET, the relevant photon energy is 511 keV from positron–electron annihilation. In the present experiment, the Bremsstrahlung photon spectrum produced by GeV-scale electrons on a tungsten target extends to substantially higher energies, with a median around a few MeV. At these energies the Klein–Nishina differential cross-section is strongly forward-peaked, producing a narrower angular distribution than the more isotropic scattering observed at 511 keV. Despite this kinematic difference, the methodology developed here — simultaneous track reconstruction and timestamping to decompose geometric and detector-induced timing

contributions — is energy-independent and directly transferable to the 511 keV regime. Furthermore, operating at higher photon energies amplifies the forward-peaking and therefore concentrates the scattered particles within the geometric acceptance of the downstream detectors, which is experimentally advantageous for a proof-of-principle measurement with limited beam time.

The use of an accelerator-based Bremsstrahlung source is not merely a convenience but a physical necessity imposed by the measurement concept itself. The central novelty of this experiment — simultaneous tracking and timestamping of the Compton recoil electron within the same event — requires that the electron traverse the full detector chain: Scintillator t_1 (5 mm plastic), two DWC gas volumes (20 cm each), and Scintillator t_2 (5 mm plastic). At PET-relevant energies (511 keV), the maximum Compton electron kinetic energy is approximately 340 keV, corresponding to a range of order 1 mm in plastic scintillator; such an electron would be absorbed immediately and could never reach the downstream tracking detectors.

By contrast, MeV-scale Bremsstrahlung photons from GeV electrons on a tungsten target produce Compton recoil electrons with tens of MeV kinetic energy — relativistic, minimum-ionising particles that comfortably traverse the entire detection chain. Furthermore, the high instantaneous photon flux achievable with an accelerator beam is essential to accumulate the 10^3–10^4 coincidence events required for angular-binned analysis within the limited beam time, a statistical target that would be impractical with laboratory radioactive sources or medical X-ray tubes. The connection to clinical TOF-PET is therefore methodological: the $\sigma^2_{total} = \sigma^2_{geo} + \sigma^2_{det}$ decomposition technique developed here is energy-independent and directly transferable to the 511 keV regime, while the accelerator environment provides the kinematic conditions under which the technique can be experimentally validated.

2. METHODOLOGY

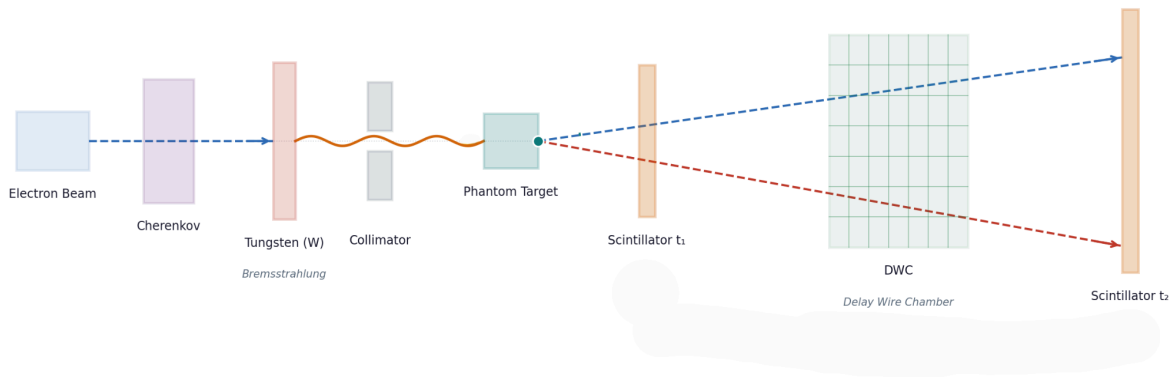

Fig. 1. *Schematic diagram of the setup layout of the experiment. Blue Line=Electron, Orange Line=Photon*

The experiment is structured as a sequential detection chain, in which each component serves a specific and well-defined role. The following subsections describe each element in the order in which particles traverse it.

2.1 Electron Beam

The electron beam constitutes the starting point of the experiment. Because the electron energy is controllable, the resulting photon spectrum is also controllable, which is necessary to reduce systematic uncertainties. The use of an electron beam, rather than a direct radioactive source, provides a variable and repeatable initial condition in terms of energy.

2.2 Cherenkov Detector

A Cherenkov detector is placed downstream of the electron beam to monitor the beam and provide an independent confirmation of each electron's passage. As the relativistic electron traverses the Cherenkov radiator, it emits a cone of light at a characteristic angle determined by its velocity. This light is collected by a

photomultiplier tube and produces a fast reference signal. In this setup, the Cherenkov detector serves as a beam tagger, providing a prompt timing signal that can be used to verify the electron beam's energy range and to cross-check the timestamps from the plastic scintillators.

2.3 Bremsstrahlung Target

When relativistic electrons decelerate in the Coulomb field of a high-Z nucleus, they emit electromagnetic radiation — a process known as Bremsstrahlung. The purpose of this target is to produce a controlled flux of x-rays for subsequent Compton scattering. In clinical PET, 511 keV annihilation photons are the relevant signal; here, Bremsstrahlung provides a tunable photon source.

The target material is tungsten (W), with a thickness of $2.5\,X_0 \approx 8.75$ mm (given W radiation length $X_0 \approx 3.5$ mm). Tungsten is selected for its high atomic number ($Z = 74$), which maximises Bremsstrahlung yield. For an electron beam energy range of 0.5–4 GeV, the optimal target thickness is between 2 and 3 X_0: thinner targets yield insufficient photon flux, while targets thicker than 3–4 X_0 cause significant self-absorption of the produced photons via the photoelectric effect and pair production. The chosen thickness of 2.5 X_0 therefore represents the optimal balance between yield and reabsorption.

2.4 Collimator

The Bremsstrahlung target produces x-rays with a wide angular spectrum. Without collimation, this would result in an irregular and poorly defined photon flux incident on the Compton target, introducing large path-length uncertainties. The collimator restricts the beam to a narrow, near-axial pencil of photons. Only photons within a small solid angle proceed to the phantom target, ensuring a well-defined interaction region and improving the accuracy of position reconstruction in the delay wire chamber.

2.5 Tissue-Equivalent Phantom Target (Compton Scatterer)

For photon energies relevant to Positron Emission Tomography, photon interactions in low-atomic-number materials are dominated by Compton scattering, lying above the photoelectric regime and below the pair-production threshold. In this interaction a photon transfers part of its energy to a loosely bound electron and is scattered with reduced energy. The kinematics of the process relate the scattering angle to the photon energy transfer through:

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_e c}(1 - cos\theta)$$

This relation enables the photon energies to be inferred from the measured scattering angle. The scattering medium used in the experiment is a tissue-equivalent phantom constructed from a plastic compound with a composition of approximately 10.5% hydrogen, 25.6% carbon, 60.2% oxygen, and 3.7% nitrogen by mass, yielding an effective electron density of approximately 3.3×10^{23} e^-/cm^3 and a physical density close to 1.04 g/cm^3. This composition approximates the radiological properties of soft biological tissue, ensuring that Compton scattering remains the dominant photon interaction mechanism in the relevant energy range. The phantom is constructed as a $3 \times 3 \times 3$ cm^3 cube.

2.6 First Plastic Scintillator (Timestamp t_1)

Plastic scintillators produce fast timing signals due to their short fluorescence decay times. As shown in Table 1 below, while LYSO(Ce) has a decay time of approximately 40 ns, plastic scintillators decay in 2–3 ns, resulting in substantially lower time jitter. Plastic scintillators are also light, sturdy, and easy to assemble, which enhances the practical feasibility of the setup.

The tasks of the first scintillator are to detect the passage of the Compton-scattered electron and to produce the first timestamp t_1, which initiates the time-of-flight measurement. To minimise energy loss and multiple scattering of the electron before it reaches the DWC, a thin scintillator tile (5 mm thickness) is used. The associated degradation of the angular resolution in the downstream DWC is estimated to be small relative to the intrinsic DWC position resolution, but will be quantified in the final analysis.

Material	Light Yield (ph/MeV)	Emission Peak (nm)	Decay Time (ns)	Density (g/cm³)
Plastic	10,000	430	2–3	1.03
LYSO(Ce)	30,000	420	40	7.10
LSO(Ce)	27,300	420	35–47	7.40
GAGG(Ce)	46,000	520	88, 258	6.63

Table 1. *Comparison of scintillator properties relevant to timing performance.*

2.7 Delay Wire Chamber (DWC)

A delay wire chamber is a gas-filled, position-sensitive detector. Charged particles traversing the gas cause ionisation; the resulting electrical signal propagates along the sense wire in both directions. The time difference between the signal's arrival at each end of the wire is used to determine the position of the ionisation event along the wire with sub-millimetre precision (typically $\sigma \approx 200$ µm).

Two DWC planes (DWC-1 and DWC-2) are placed downstream of Scintillator t_1, separated by a known longitudinal distance. The DWCs are not used for timing in this experiment. Their role is to reconstruct the two-dimensional transverse position (x, y) of the scattered particle in each plane, determine the Compton scattering angle θ from the reconstructed trajectory vector relative to the beam axis, and compute the geometric path length L between the phantom target and the second scintillator.

The scattering angle is reconstructed as follows: the hit positions in DWC-1 and DWC-2 define a three-dimensional trajectory vector. The polar angle of this vector with respect to the beam axis yields the Compton scattering angle θ. The expected angular resolution, determined by the DWC spatial resolution (~200 µm) and the inter-chamber separation, is estimated to be better than 0.5° for the angular range of interest ($\theta < 30°$). Without this positional information, it would be impossible to determine what fraction of the observed timing-distribution broadening arises from geometric path-length variation as opposed to intrinsic detector jitter.

2.8 Second Plastic Scintillator (Timestamp t_2)

The second plastic scintillator records the arrival of the scattered electron and produces the second timestamp t_2. The time-of-flight is then: $\Delta t = t_2 - t_1$

This time difference is recorded for each event and compiled into a histogram. The full width at half maximum (FWHM) of this distribution represents the integral timing resolution of the system. This is the central observable of the experiment.

2.9 Trigger and Data Acquisition

The coincidence trigger is formed from the logical AND of discriminated signals from Scintillator t_1 and Scintillator t_2. Both scintillator signals are fed through leading-edge discriminators into a NIM coincidence unit; the output of this unit gates the data-acquisition system. The Cherenkov detector signal is recorded but not included in the trigger, serving instead as an offline beam-quality tag and timing cross-check.

For each triggered event, the following signals are digitised: the Scintillator t_1 and t_2 timing signals via a multi-hit TDC (time-to-digital converter) with a bin width of ≤25 ps, providing the Δt measurement; the Cherenkov timing signal via the same TDC; the pulse heights from both scintillators via a charge-integrating ADC, providing energy-deposition information for Compton kinematics cross-checks; and the four DWC timing outputs (two per chamber, for x and y coordinates) via the TDC. The standard T9 beamline DAQ infrastructure (NIM/VME crate with CAEN modules) will be used. Event data are read out through a VME controller and written to disk for offline analysis.

2.10 Background Rejection

Not all events triggering the scintillator coincidence will correspond to clean Compton-scattered electrons. Potential background sources include direct beam fragments leaking through the collimator, electron–positron pairs from photon conversions in or near the phantom, and multiply scattered particles with degraded angular correlation. Several selection criteria will be applied offline to enrich the Compton signal:

An energy-deposition window in Scintillator t_1 consistent with the expected Compton electron energy at the reconstructed angle, rejecting minimum-ionising particles from the primary beam and high-energy pair-production leptons.

A fiducial cut on the DWC-reconstructed transverse position, requiring the trajectory to originate within the phantom volume.

An angular consistency cut, requiring agreement between the scattering angle reconstructed from the DWC track and the angle predicted from the Compton formula using the measured energy deposition.

The residual background fraction will be estimated from the sideband regions of these distributions and propagated as a systematic uncertainty on the extracted σ_{geo} and σ_{det} values.

2.11 Analysis Method

The path length L of the scattered electron between the phantom target and Scintillator t_2 is computed event-by-event from the DWC-reconstructed transverse position (x, y) and the known longitudinal separation Δz between the target and detector planes, according to $L = \frac{\Delta z}{cos\Theta}$

The standard deviation of the resulting L distribution across all recorded events directly yields the geometric timing contribution as $\sigma_{geo} \approx \frac{\sigma(L)}{c}$. To isolate this contribution from the intrinsic detector jitter, events are binned by reconstructed scattering angle θ. Within each narrow angular bin, path-length variation is suppressed, and the FWHM of the Δt distribution within that bin converges toward the detector-only limit σ_{det}. The geometric component σ_{geo} is then extracted by subtracting σ^2_{det} in quadrature from the full-sample FWHM, yielding the decomposition $\sigma^2_{total} = \sigma^2_{geo} + \sigma^2_{det}$ entirely from experimental data, without reliance on simulation input at the analysis stage. The angular bin width will be optimised to balance the suppression of geometric variation within each bin against the statistical precision of the per-bin FWHM extraction.

3. PARAMETERS TO BE MEASURED

The following quantities will be measured and recorded for each event:

- Δt: the time difference between the two scintillator timestamps t_1 and t_2, giving the time-of-flight of the scattered electron
- θ: the Compton scattering angle, derived from two-dimensional track reconstruction using the DWC
- (x, y): the transverse position of the scattering event within the phantom target, as reconstructed from the DWC
- L: the geometric path length of the scattered electron between the phantom target and the second scintillator, computed from the DWC and detector geometry
- FWHM of the Δt distribution: quantifying the integral timing resolution of the full system
- Energy deposition in each plastic scintillator, to be correlated with the predicted Compton electron energy

By combining timing measurements with track reconstruction, it will be possible to separate the geometric contribution σ_{geo} from the intrinsic detector contribution σ_{det} to the overall timing resolution.

4. PRELIMINARY SIMULATION RESULTS BASED ON GEANT4

A Geant4 Monte Carlo simulation of the full beamline was conducted prior to submission to validate the physical model and provide quantitative predictions against which the beam run results will be compared. The simulation models a 500 MeV electron beam incident on a 2.5 X_0 tungsten Bremsstrahlung target, with the resulting photon flux collimated onto the cylindrical phantom Compton target described in Section 2.5.

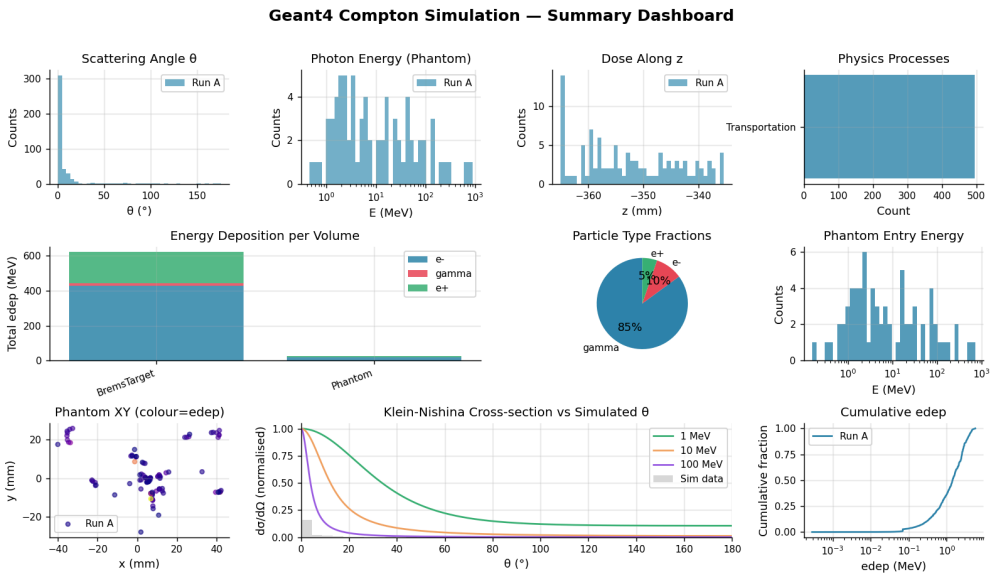

Fig. 2. *Geant4 simulation summary dashboard. Top-left: simulated Compton scattering angle distribution, consistent with Klein–Nishina forward-peaking. Top-right: Bremsstrahlung photon energy spectrum at the Phantom. Middle: energy deposition per volume. Bottom-centre: comparison of simulated θ distribution with normalised Klein–Nishina differential cross-sections at 1, 10, and 100 MeV.*

The angular distribution of simulated Compton photons in the Phantom volume is strongly forward-peaked: all events scattered within 30° of the beam axis, with 81% within 10° and a mean scattering angle of $\bar{\theta} = 6.9°$ ($\sigma = 4.4°$). This is consistent with Klein–Nishina predictions at high photon energies and confirms that the DWC and Scintillator t_2, positioned downstream along the beam axis, will intercept the vast majority of scattered particles without requiring wide angular acceptance. The Bremsstrahlung photon spectrum spans 0.4–97.7 MeV with a median of 3.8 MeV, and when the simulated photon energies are compared to the Klein–Nishina kinematic boundary at the beam-peak energy, the simulation correctly populates the sub-boundary region, confirming internal physical consistency.

Energy deposition in the Phantom volume is exclusively from electrons, with no gamma or positron contribution. This confirms that Compton scattering is the dominant interaction mechanism in the phantom target as intended, with negligible contamination from photoelectric or pair-production processes, as expected for a low-Z material at these energies.

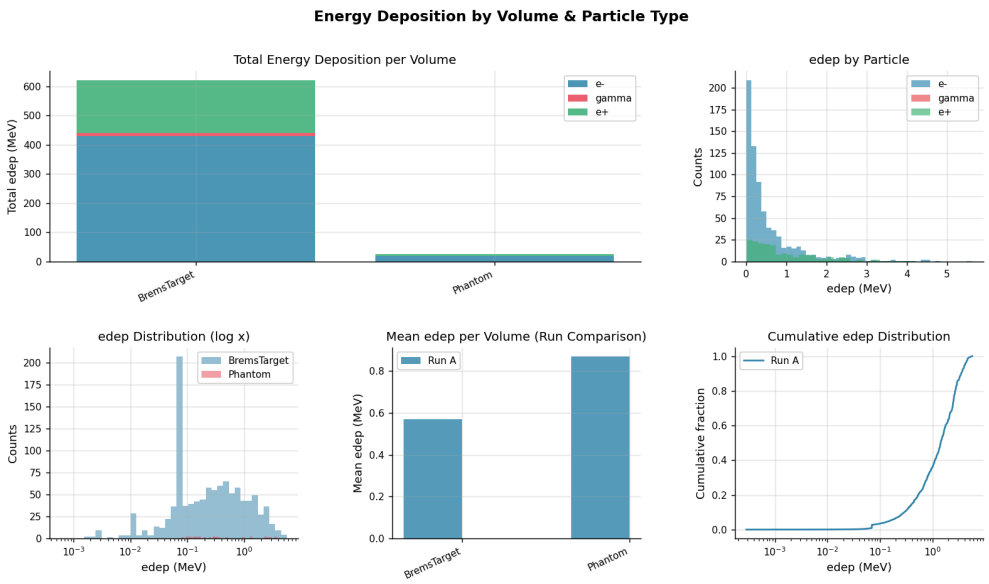

Fig. 3. *Geant4 simulation of energy deposition by volume and particle type. Top-left: total energy deposition per volume broken down by particle species, showing that the BremsTarget receives the dominant energy deposit (~620 MeV) while the Phantom receives a comparatively small fraction (~20 MeV). Electron (e^-) deposition dominates in both volumes, with negligible gamma and positron contributions in the Phantom, confirming that Compton scattering is the dominant interaction mechanism in the target as intended. Top-right: energy deposition histogram by particle type, showing the expected falling spectrum with electron events concentrated below 1 MeV. Bottom-left: log-scale distribution of energy deposition across both volumes, illustrating the broad dynamic range of deposited energies. Bottom-centre: mean energy deposition per volume comparison, showing the Phantom mean (~0.85 MeV) exceeds the BremsTarget mean (~0.57 MeV) on a per-event basis despite lower total deposition. Bottom-right: cumulative energy deposition distribution, showing that approximately 80% of events deposit less than 1 MeV, relevant to the scintillator detection threshold and trigger efficiency estimates.*

Using the simulated transverse hit spread in the phantom volume (radial $\sigma \approx 24.9$ mm) and the phantom-to-scintillator longitudinal separation ($\Delta z \approx 232$ mm), the geometric path-length variation translates to a timing spread of $\sigma_{geo} \approx 83$ ps. Since plastic scintillators achieve intrinsic timing resolutions of 200–300 ps FWHM, this geometric contribution represents approximately 28–41% of the total timing resolution budget — non-negligible in magnitude and directly measurable by the proposed coincidence setup. These results provide the quantitative motivation for the experiment and constitute a falsifiable prediction for the beam run. A higher-statistics simulation run ($>10^5$ primary events) is currently in preparation to provide more refined predictions for the angular-binned analysis and background contamination estimates.

Collectively, these simulation results validate three core assumptions underpinning the experimental design: that Compton scattering is the dominant interaction mechanism in the phantom, that the forward-peaked angular distribution is well-contained within the geometric acceptance of the downstream detectors, and that the geometric timing contribution σ_geo is sufficiently large relative to σ_det to be resolved by the proposed coincidence setup. The simulation therefore confirms that the proposed measurement is not only physically motivated but experimentally tractable within the available beam time and detector configuration.

	Cherenkov	Brems. Target	Collimator	Phantom	Scint.-1	DWC-1	DWC-2	Scint.-2
Material	CO_2	W	Pb	Tissue-eq. plastic	Plastic scint.	Ar gas	Ar gas	Plastic scint.
Composition	—	Z=74	Z=82	H:10.5% C:25.6% O:60.2% N:3.7%	—	—	—	—
Thickness	10 cm	0.875 cm	5 cm	3 cm	0.5 cm	20 cm	20 cm	0.5 cm
Density	—	19.3 g/cm³	11.3 g/cm³	1.04 g/cm³	1.03 g/cm³	—	—	1.03 g/cm³

Table 2. *Beamline Component Specifications*

5. SCIENTIFIC INTENTION OF THE EXPERIMENT

The primary scientific goal of this experiment is to quantify the contribution of Compton scattering geometry to the integral timing resolution of a TOF-based detection system. Specifically, we aim to determine how much of the observed broadening in the Δt distribution arises from the variation in geometric path lengths of scattered electrons, as opposed to intrinsic detector timing jitter.

This distinction is of direct relevance to the design of TOF-PET systems used in oncological imaging. In clinical PET, improved timing resolution translates to a better signal-to-noise ratio and more precise localisation of the annihilation vertex, ultimately enabling more targeted radiotherapy with reduced dose to surrounding healthy tissue.

By correlating the reconstructed scattering angle and path length (from the DWC) with the measured time difference (from the scintillators), we will construct a dataset that allows the geometric and detector-induced components of the timing resolution to be disentangled. The results are expected to provide experimentally grounded constraints for Monte Carlo simulations and detector optimisation studies relevant to future TOF-PET instrumentation.

Furthermore, this experiment provides a hands-on investigation of the interplay between Compton kinematics and detector performance — a relationship that is central to the ongoing development of next-generation PET systems targeting sub-100 ps timing resolution.

6. FEASIBILITY AND EXPECTED STATISTICS

The proposed measurement is feasible within the constraints of a standard BL4S beam run. All required instrumentation — delay wire chambers, plastic scintillators, and a Cherenkov beam tagger — is available as standard equipment at the T9 beamline, and the tissue-equivalent phantom requires no specialised fabrication.

Based on the expected Bremsstrahlung photon flux and Compton cross-section in the relevant energy range, a dataset of approximately 10^3–10^4 coincidence events with DWC-reconstructed tracks is a realistic target within the allocated beam time. For the angular-binned analysis, the bin width will be chosen to ensure a minimum of ~100 events per bin, which is sufficient for a statistically meaningful FWHM extraction. With a 2° bin width over the expected 0–30° range, a total sample of ~1500 events provides approximately 100 events per populated bin, constituting the minimum from which a reliable $\sigma_{geo}/\sigma_{det}$ decomposition can be extracted.

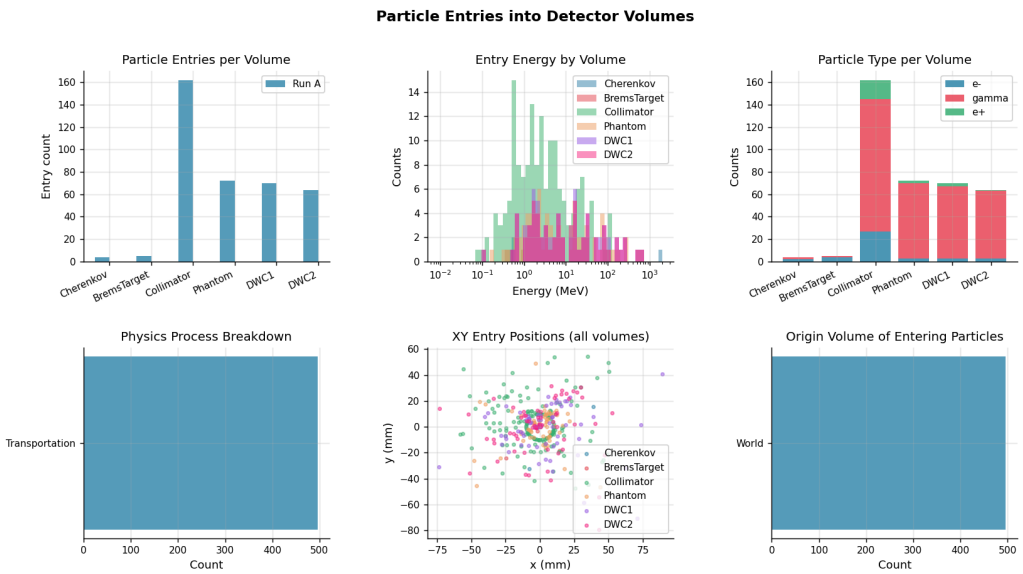

Fig. 4. *Geant4 simulation of particle entries into detector volumes. Top-left: total entry counts per volume, showing the Collimator receives the highest particle flux (~160 entries) while the Phantom, DWC1, and DWC2 each receive 65–75 entries, consistent with the 10^3–10^4 coincidence event target within the allocated beam time. Top-centre: entry energy spectrum by volume, showing particles arriving at the DWCs and Phantom predominantly in the 1–100 MeV range, well above the scintillator detection threshold. Top-right: particle type breakdown per volume, confirming that gamma photons dominate entries into the Collimator and downstream volumes, with a small but measurable electron and positron fraction relevant to background rejection. Bottom-centre: reconstructed XY entry positions across all volumes, showing the beam is well-contained within a ~150 mm transverse radius, confirming that the downstream detectors intercept the relevant particle flux without requiring an extended geometric acceptance.*

Should this target be met, the experiment will yield the first directly measured, track-assisted decomposition of geometric and detector-induced timing contributions for a Compton-based TOF geometry — a reference result of direct relevance to next-generation TOF-PET detector design targeting sub-100 ps total timing resolution.

7. TEAM MOTIVATION

Beyond the scientific objectives, we are motivated by the opportunity to carry out real measurements at a world-class accelerator facility and to analyse the data ourselves. Experiencing how experiments are planned and executed at CERN, and how detector systems are used to probe the fundamental building blocks of the universe, will deepen our scientific understanding and inspire our future academic careers.

8. ACKNOWLEDGEMENTS

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